



**Politecnico
di Torino**

Department of Electronics and Telecommunications
Bachelor degree in Electronic Engineering

11APMNX - Dispositivi Elettronici

Lab assignment 2: Silicon PN Junction

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Intro duction

The second laboratory assignment is composed of different exercises to be solved autonomously using the PADRE tool on nanoHUB. In particular, it is focused on the analysis of a silicon PN junction under different working conditions and at different abstraction level. The starting point is the simulation of a silicon PN junction in equilibrium condition, then a bias voltage must be applied in order to analyse the device behaviour out of equilibrium.

In order to get an evaluation, you **must** write a brief report in which you insert all the main outcomes obtained from the simulations together with an explanation and a comparison with the theoretical results. At the end of the report, you also have to insert all the PADRE input files necessary to perform the simulations. In the report follow the same structure of the assignment: highlight the answers by writing close to them the number of the corresponding question.

The report must be submitted in PDF format on the “Portale della Didattica > Elaborati”.

The report of this second assignment **must** be submitted within the **20th January 2026**.

For each exercise, specifications will be calculated considering some of your personal information: name, surname e student ID number. From now on:

- **#n**: is the number of characters of your name;
- **#s**: is the number of characters of your surname.

For example, considering “Mario Bianchi”, #n = 5 and #s = 7.

IMPORTANT: Please write in the report the specifications that you obtain using your personal information. Report both #n and #s and also the derived quantities like e.g., doping values.

IMPORTANT: Always specify in your answers the unit of measurement of all the quantities, in both graphs and formula results. The measurement units should be written in print (i.e., block letters), NOT in italic; a space should be present between the number (physical quantity value) and the measurement unit.

The report can be either in English or in Italian according to your preference.

Report also the PADRE and LTSPICE input files you develop to do the assignment. They may be reported within the report or attached in a .zip file and uploaded together with the report.

Important Remark: To assign you the data of the exercises, we will make use of the mod operation. It is the remainder after dividing one number by another. Example: $100 \bmod 9$ equals 1. Because $100/9 = 11$ with a remainder of 1.

EXERCISE A: PN Junction physical simulation

Consider the silicon P-N junction of Figure 1 at 300 K.

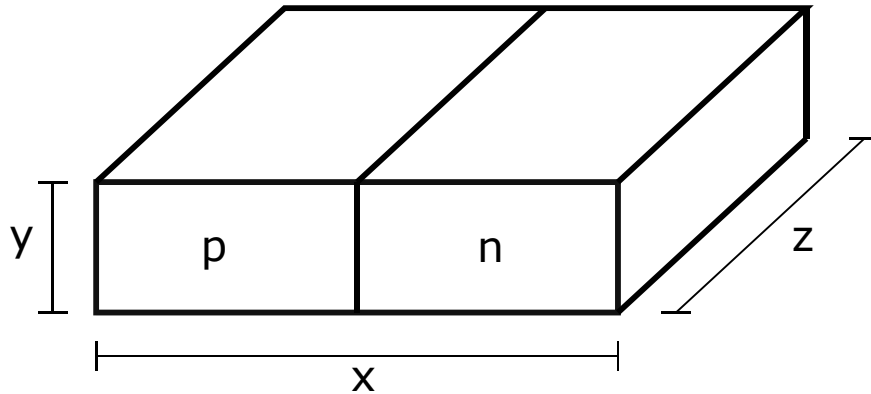


Figure 1: Dimensions of the silicon sample.

The doping concentrations of both regions is uniform and defined as follows:

- **p-type region:** $N_A = a \cdot 10^{16} \text{ cm}^{-3}$, where $a = \#s \bmod 9 + 1$;
- **n-type region:** $N_D = b \cdot 10^{16} \text{ cm}^{-3}$, where $b = \#n \bmod 9 + 1$.

The function *mod* is the **modulo operation**, which returns the remainder of a division, where the number before the operator is divided by the one after the operator. The electron and hole lifetimes are equal to $0.2 \mu\text{s}$. The device dimensions are calculated in micrometers (μm) as follows:

- **x:** is $(\#n + \#s)$ times the sum of the last 4 digits of your student ID number;
- **y:** is $\#s$ times the sum of the first 2 digits of your student ID number;

The junction is placed in the middle of the device at $x/2$.

For example, considering “Mario Bianchi” and “s123456”, $x=216\mu\text{m}$, $y=21\mu\text{m}$ and the junction is placed at $x=108 \mu\text{m}$.

Questions

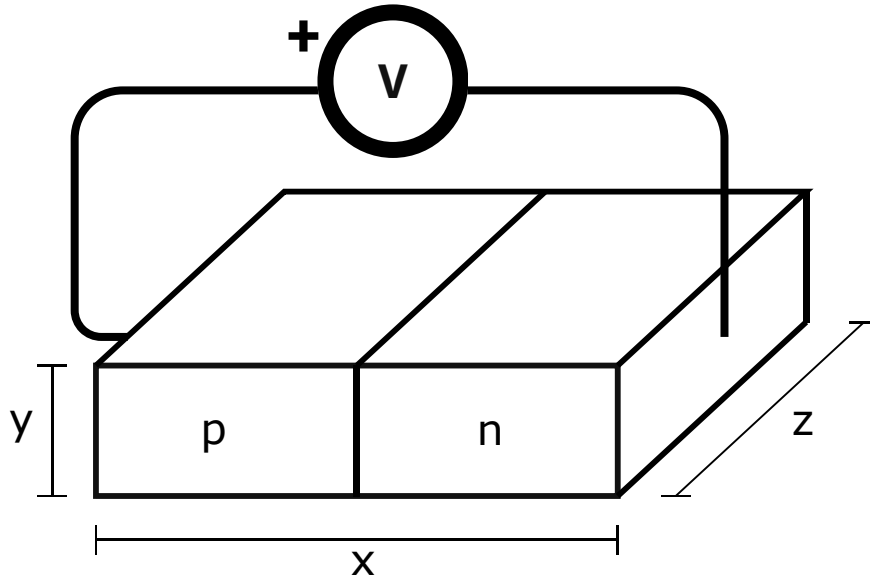
Q A.1: Plot the band diagram obtained from PADRE in equilibrium conditions and comment on the behaviour of the energy bands. Report the value of the built-in potential V_{bi} and compare it with the analytical formula.

Q A.2: Plot the net space charge concentration from PADRE in equilibrium conditions. Calculate the width of the depletion region of both sides and evaluate the profile of the net charge concentration using the analytical formula. Compare the results and comment on them.

Q A.3: Plot the electric field from PADRE in equilibrium conditions. Evaluate the profile of the same quantity using the analytical formula and compare the results, commenting on them.

EXERCISE B: PN Junction out of equilibrium

Consider the same structure as the Ex. before:



Questions

Q B.1: Apply a bias V_a (rounded to the first decimal digit) evaluated as:

$$|V_a| = \left[e^{-\left(\frac{\#n}{\#n + \#s}\right)^2} - (1 - V_{bi}) \right] V$$

where the applied voltage V_a is positive if $(\#n + \#s) \bmod 2 = 0$, otherwise the applied voltage is negative. Then:

1. Plot the band diagram obtained from PADRE and comment on the behaviour of the energy bands and highlight the differences with respect to the equilibrium conditions.
2. Plot the space charge concentration from PADRE. Calculate the width of the depletion region of both sides and evaluate the profile of the net charge concentration using the analytical formula. Comment on the behaviour and highlight the differences with respect to the equilibrium conditions.

3. Plot the electric field from PADRE. Evaluate the profile of the same quantity using the analytical formula and compare the results. Comment on the behaviour and highlight the differences with respect to the equilibrium conditions.

4. Plot the current densities (Electron Current, Hole Current and Total Current) obtained from PADRE and comment on its behaviour in the PN junction. Calculate the same quantities using the analytical formulas and compare the results.

(For an effective comparison between the equilibrium and the out-of-equilibrium values, a superposition of the two graphs is suggested)

Q B.2: Modify the template to obtain a IV characteristic, with voltage values that go from $-c/5$ to $+n/5$. Plot the characteristic to a single graph and comment on it.

(Hint: to do so you will need to combine two simulations: one for a forward bias and one for a reverse bias. If PADRE doesn't handle a voltage that is too high, scale it down and report it)